

Personify App

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Session 2021-2025

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Personify App

A project submitted to the
Department of Computer science

In

Partial Fulfilment of the Requirements for the
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DECLARATION

We hereby declare that this project titled “**Personify**” is not copied out from any other source. This authentic work done by signatory, in practical achievement for the degree “Bachelor of Science in Computer science” at Computer science Department, Lahore Garrison University, Lahore. Besides, it is declared that we have designed this APP or project report completely on the basis of our own intention under the kind leadership of our internal supervisor Mam Ramesha Rehman.

All the analysis, design & system development have been accomplished by the signatory. Moreover, this project has not been submitted to any other college or university.

Muhammad Adeel _____

Munawar Ihsan_____

DEDICATION

I have devoted this piece of work to First of all, ALLAH Almighty who honored us with the knowledge and courage to finish this task with grace. Secondly, my loving supportive family whose prayers, suggestions and ongoing encouragement helped a great deal in reaching this destination. Last but not least, to the Lahore Garrison University and Particularly, the department of computer science.

ACKNOWLEDGEMENTS

First and foremost uncountable thanks to ALMIGHTY ALLAH who made us worthy of all praise who guide us in times of crises. Secondly all greetings are offered to our dearest PROPHET MUHAMMAD (PBUH) who give us the power to identify our creator and whose life is the ultimate guide for us.

Throughout this project, we were assisted, aided or practically helped by numerous individuals and it is a bliss to pay homage to their intentions. Our sincerest prayers to our parents, brothers and sisters and to all the members of our family for their support and moral motivation. Our success is only because of prayers or support of our mothers who actually loves us and pray for us in every second of our life. We are highly indebted to all the members of our family, senior's peers and we also feel highly indebted to our project guide Mam Ramesha Rehman for having given us accurate research-based views, advises, support and guidance throughout the project and in the academic year.

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LIST OF ABBREVIATION

Abbreviation	Full Form
App	Application
Api	Application Programming Interface
DB	Data base
DFD	Data Flow Diagram
GUI	Graphical User Interface
HTTP	Hyper Text Transfer Protocol
UI	User Interface
UX	User Experience
VS	Visual Studio Code

ABSTRACT

Current affairs taking into consideration many cases include child kidnappings and smuggling. They separated many parents and their children. Many people have accidents and sudden deaths. When anyone found their body, they didn't recognize where the person belongs too. Many criminals are roaming around us. Who has a criminal record? So, you don't have any solution to identify them. The solution to all these problems is PERSONIFY. PERSONIFY is a project that consists of two apps. One of them is PERSONIFY Admin for the authorities, who authenticate the case of an individual and input that case information into the database. Another is for individuals referred to as PERSONIFY Client, where individuals simply snap a photo of the victim in real time and send it to PERSONIFY, so if such a person is available via the database, it will retrieve the record of that individual.

Keywords: Kidnappings, Smuggling, Recognize, Criminal Record.

INTRODUCTION

1.1. Background

Normally, the client will be requested to register in order to login the application. When the client selects the image to upload to the server, the information will be received by the client when there is a match in the database. The admin will also have to login in his side of the app and be required to fill out a form and upload the picture in to the database to keep a record.

1.2. Overview

This App Utilizes the potential and benefits of face recognition based on Face Landmark detector. PERSONIFY is based on a concept to maintain the security, convenience and provides improved facial expression analysis. Anyone with even a minute knowledge of usage of android applications can avail themselves of this purpose.

1.3. Objective

There are many reasons for an application, in this case utilizing the innovation of facial recognition algorithm, Authorities' can accurately recognizing faces, Identifying individuals, improved security and improved convenience. At this time this application will be available in a specific area.

1.4. Gantt Chart

Gantt Chart of our APP is given below in table 1.

Table 1- Gantt Chart.

Activities	Week/Semester
Front End	Week 7 & 9 (7th)
Data-Base	Week 10, 11 & 12 (7th)
Wish list	Week 13 & 14 (7th)
APP Security	Week 15 (7th) Week 1 (8th)
Add to Cart	Week 2, 3 & 4 (8th)
Login details	Week 5, 6 & 7 (8th)
Home Screen	Week 9, 10 & 11 (8th)
History Creator	Week 12 & 13(8th)
Documentation	Week 7 (7 th) – Week 13 (8 th)

1.5 Report Organization

This report is organized chapter by chapter, as each chapter explains the important tasks involved in carrying out the project. We have included a report for every chapter, as detailed below.

Chapter 1: also covered the background and an overview of App, as we discussed earlier. The chapter discussed Gantt chart which lists the evaluation timeline.

Chapter 2: Looks at what it means to define the problem that the project was created to solve.

Chapter 3: Software Requirements Specification, with its functional and nonfunctional parts, was introduced.

Chapter 4: Provides information on technology and methods essential for the project process.

Chapter 5: Covers system structure and performance by means of diagrams. The lesson covers every aspect of the ER.

Chapter 6: The processes of testing, creating test plans and maintenance measures for the project are given.

Chapter 7: Describes how to assess study results.

Chapter 8: This chapter concludes the entire working and post work of the project.

LITERATURE REVIEW

Facial recognition and identification is one of the greatest and effective application invented by Artificial Intelligence. Reported as "The Most Dramatic Application Of Artificial Intelligence," this technology is employed in a wide variety of high-stakes industries such as security, surveillance, personal identification and law enforcement. Its impact stretches from the public sector into the private, bringing identity-related responsibilities for institutions and governments to new levels of efficiency and accuracy.

An important factor in the achievements of facial recognition is the combination of deep-learning algorithms. Deep learning is changing the way machines see and understand faces. Now systems can learn from large data sets thanks to advanced neural networks and can detect and identify faces under challenging environmental conditions. These conditions can range from partial occlusion (where other parts of the face are occluded, but not over the face), low resolution input or even time-related changes in faces due to aging, or expression variation). Regardless of these limitations, contemporary face identification systems can achieve an incredible level of accuracy, which makes them very reliable instruments in high risk environments.

In recent years, facial recognition has expanded beyond its heritage applications to mobile and cloud-based solutions. The mobility and user-convenience of mobile devices and the massive storage and computing capability being provided by the cloud, gains opportunities for the real-time, location independent recognition tasks. These developments, moreover, have enabled the creation of applications that are not only powerful and accurate, but also practical and user-friendly for both governmental and public users.

Problem Definition

Current undertakings thinking about many cases incorporate kid kidnappings and pirating. They isolated many guardians and their youngsters. Many individuals have mishaps and abrupt passing. At the point when anybody found their body, they didn't perceive where the individual should be too. Peoples around us have criminal records, capturing youngsters and selling their organs on the black market. So the government departments and your brothers and sisters need their people's help. It's the responsibility of every citizen to play a key role in this activity.

Chapter 4

Software Requirement Specification

4.1. Purpose

With Deep Learning leading its features, our Personify App is built to help with the search for missing people. This app was developed to help locate missing persons so their loved ones can be brought together and have the opportunity to move on. Officials in law enforcement use it to assist with crime investigation and the apprehension of offenders. Because of this application, you can submit all information on a missing person by clicking just one button and find out who is closest to the person you lost.

Furthermore, it can help police identify suspects by comparing images or video footage of a person's face to a database of known criminals or suspects. This can be particularly useful in cases where a suspect is wearing a mask or hat or is otherwise difficult to identify through traditional means. By using face recognition, police can quickly identify and apprehend suspects who may pose a threat to public safety, such as individuals involved in terrorism, organized crime, or other dangerous activities. It can be used to help solve crimes by identifying suspects or witnesses captured in surveillance footage.

4.2. Revision History

No revision history.

4.3. Document Conventions

This report employs the following delegations as appear in List of Abbreviations [2].

4.4. Intended Audience and Reading Suggestions

According to that report, both the developer and user should make some adjustments [3].

- Developers use this documentation to get an overview of the framework used, issues with performance and function and processes and data handling in the App.
- User will check this guide to discover what main functions the APP has as determined by its user interface.
- Using paper is recommended when engineering, domain expert, developer and project manager needs arise.
- It is strongly advisable to first look at the Vision Document when wanting to understand the project before this document. You are provided with proper guidance and information about the app through the document.

4.5. Product Scope

The aim of the project is to put forward a solution for the missing people. It can locate and reunite the missing persons with their loved ones. As well as help to solve crimes and bring perpetrators to justice. It can not only locate and match already people from the database. But is also capable to detect the person from a live CCTV cam feed. Perks of this product are:

- **Efficiency:** A missing person finder app can help locate missing persons more quickly and efficiently than traditional methods, by analyzing large amounts of data and identifying potential matches automatically.
- **Wider reach:** An app can be accessed by anyone with a smartphone, potentially increasing the number of people who can help search for missing persons.
- **Constant availability:** An app can be accessed at any time, making it easier for people to report sightings or provide other information about missing persons.

- **Alerts and notifications:** An app can send alerts and notifications to users when there are updates or new information about missing persons, helping to keep people informed and engaged in the search.
- **User profiles:** An application can enable users to establish profiles with details about themselves and their missing loved ones, and in this way, enable individuals to give appropriate information and remain affiliated with the search.
- **Database of missing persons:** There can be a database of missing persons within an app, and it becomes convenient for individuals to search for and retrieve information regarding a particular case.

4.6. Overall Description

4.6.1. Product Perspective

Simple things are changing and ongoing change in the digital era new concept, ideas and products get introduced one after another. Deep learning is a form of machine learning that consists of training artificial neural networks on huge data sets. It is usually employed for operations that involve the capacity to learn and adapt, e.g., image and speech recognition, natural language processing, and predictive modeling.

Deep learning would be applicable in the case of a missing person finder app to enhance the effectiveness and accuracy of a missing person search. A deep learning model, for instance, would be trained on a huge dataset of missing person images and utilized to scan images and video clips to determine potential matches. This would assist police and other agencies to more rapidly and efficaciously find missing individuals and reunite them with their families.

4.6.1.1. Free login account

To use the extra information, the customer will need to create a free log-in account.

4.6.1.2. Categories of interests:

It will include the various features:

- **Accuracy:** The accuracy of the face recognition system is paramount in ensuring identification of the right individual. This category would emphasize the capacity of the system to accurately detect and match faces, especially under difficult conditions like low light levels, occlusion, or facial expression variations.
- **Speed:** The speed of the face recognition system is significant in applications that need real-time processing, e.g., video surveillance. This category would deal with the capacity of the system to handle a large volume of data in a fast and efficient manner..
- **Usability:** Ease of use of the face recognition system is necessary for guaranteeing that users can readily and efficiently interact with the system. Usability would cover the user interface, system design, and integration ease with other systems.
- **Privacy and Security:** The privacy and security of the face recognition system are vital in ensuring that user data is protected and cannot be misused. This category would center around initiatives like encryption, access control, and data retention policy.
- **Bias and fairness:** The potential for bias and fairness issues is a significant concern in face recognition technology, particularly for law enforcement applications. This category would focus on ensuring that the system is designed and implemented in a way that mitigates the risk of bias and ensures fairness for all individuals.

4.6.2. Product Functions

- Only Android phones can be used with these. In video or photo, the dlib face detector will detect the location of faces.
- Using the dlib face landmark detector algorithm, this algorithm locates facial features such as eyes, nose and mouth.
- Face recognition involves finding similarities between a person's face and faces kept in a database.
- Design of the user interface should make it possible for users to quickly use the system by viewing their data, adding and removing images or video and interacting with the app.

- Ensuring vast data on human faces is managed in a safe and orderly manner, using data encryption, managing who can use it and setting policies on keeping it.
- Processing Bulk Data: Easy to process a rough of data and add more resources when needed. We have discussed the main technical requirements here.
- Making the system work more efficiently and faster by using caching, working on different processors and optimizing algorithms.
- Using other spy cameras makes it possible to identify faces in real time.
- Protecting data and the correct use of the system: Using user authentication, encrypting data and ensuring proper access to it.
- Design and develop techniques to reduce the chance of bias and ensure fairness to all, primarily in law enforcement.

4.6.3 User Classes and Characteristics

1. End-users:

- Features: broad public, non-expert, non-technical, diverse age group, gender, ethnicity, and socio-economic groups.
- Applications: Use the face recognition app for various purposes, e.g., to unlock a device, gain entry to a secure area, identify individuals, or make photos look beautiful.

2. Developers:

- Features: technical, software developers, computer scientists, researchers, and domain professionals in machine learning, computer vision, or artificial intelligence.
- Usage: Utilize dlib face landmark to execute, experiment with, and refine face recognition algorithms, or develop new applications based on the library's face detection and landmarking capabilities.

3. System administrators:

- Characteristics: IT professionals, security professionals, and administrators responsible for deploying and operating the face recognition application.
- Usage: Install, configure, and manage the application's security settings, access controls, user authorizations, and system integration.

4. Data scientists:

- Role: data analysts, statisticians, and machine learning specialists who are responsible for training, testing, and tuning face recognition algorithms.
- Application: use dlib face landmark to preprocess and analyze face images, train and test machine learning models, and validate the accuracy and stability of the algorithms.

5. Business analysts:

- Characteristics: business strategy, market analysis, and customer interaction non-technical specialists, tasked with examining the relationship behaviors of the application and the discovery of potential revenue streams.
- Usage: Use user behavior, usage patterns, and customer comments to determine emerging features, applications, or monetization opportunities.

4.6.4. Operating Environment

PERSONIFY in order to use the app, you must have Windows and a camera installed on your computer. Write your application in Python because it is compatible with dlib face landmark.

4.6.5. Design and Implementation Constraints

The Tool is actually an important Python programming language application called PERSONIFY. It must follow the rules around data protection and privacy that relate to its users. For instance, face information has to be kept safely, all exchanged data must be encrypted and data should stay anonymous when necessary. Designers should normally be aware that the restrictions of the language and its libraries can be quite common in application creation. Supporting libraries, including the dlib face landmark, need special versions for the application to function. English according to International Standard (UK) is used for the product and it runs in accordance with the laws and rules in Pakistan. As a result of these existing laws and instructions, the data from customers continues to be protected.

4.6.6. User Documentation

Overview of the application in terms of what it does, its functionality, and system specifications. This could also cover an explanation on how to install and configure the application.

4.6.7. Assumptions and Dependencies

Researchers made the following assumptions and dependences when studying the PERSONIFY App.

- Thoughts on how Face Recognition functions.
- Guesses about the way users will navigate your site.
- The need for using Third-Party Libraries and Frameworks.

Using high-quality data is often important for creating an application that works and is accurate. Things that were assessed were the number and range of faces appearing in the training pictures, how easy the images allowed for face identification and possible biases present in the data.

4.7. External Interface Requirements

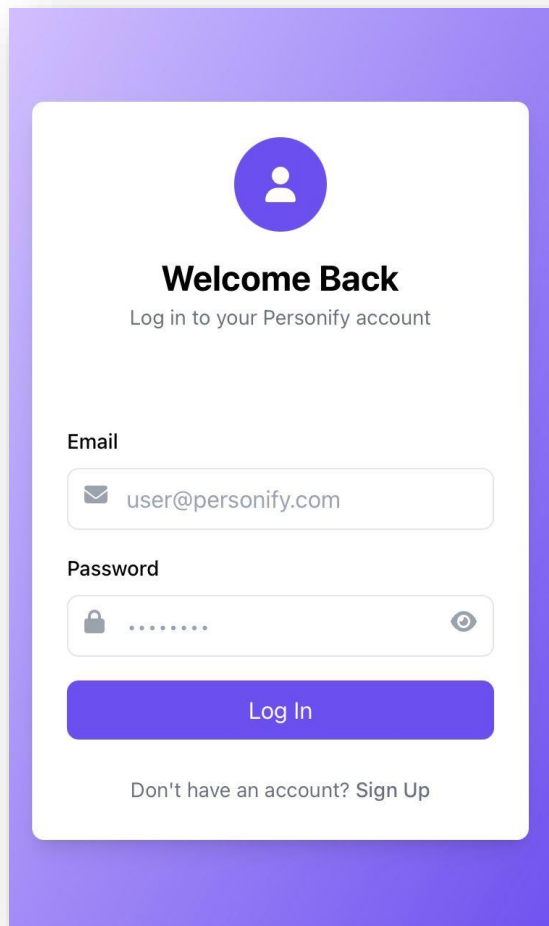
4.7.1 User Interfaces

When user use PERSONIFY Admin APP for the very first time should see splash activity page when user opens the application, see in Figure 1.

After splash screen the user is navigated to login screen, see in Figure 2. From login screen user can navigate to forget password screen in case the user forgets the password, See in Figure 4. From login screen user can also go to Signup screen if in case of first impressions, See in Figure 3. Once signed in the user is brought to a screen where the user enters the person details, fills a form and uploads the image, See in figure 5, 6.

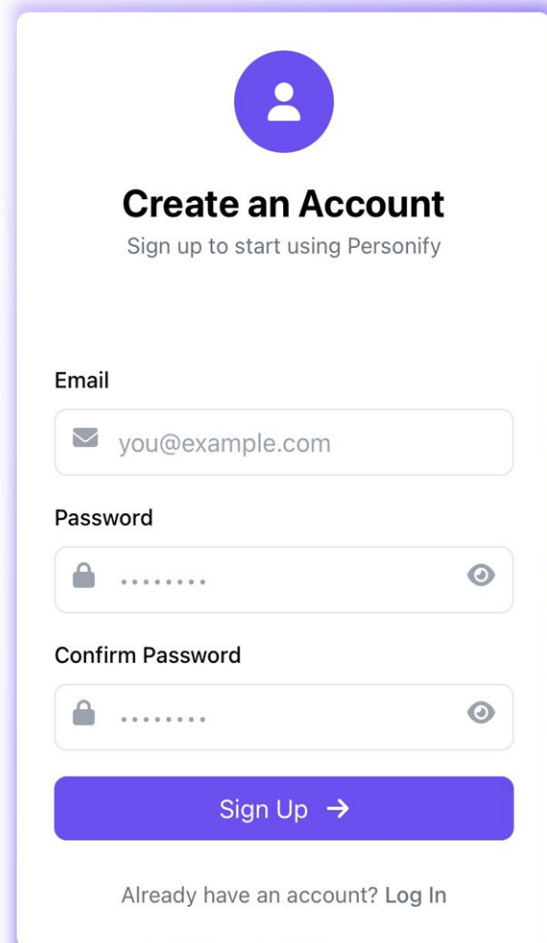
In the PERSONIFY Client App User Gets to see the Login Screen, See in Figure 7. After Signing in the user is navigated to a screen where the user has to select an image and upload it. Upon

getting a match from the server the gets to see the detailed record by clicking the details button,
See in figure 8.



The login screen features a purple header and footer. The main content area is white and contains a purple circular icon with a white person silhouette. Below the icon, the text "Welcome Back" is displayed in bold, followed by "Log in to your Personify account". There are two input fields: "Email" with the placeholder "user@personify.com" and "Password" with a masked password ".....". A purple "Log In" button is positioned below the password field. At the bottom, there is a link that says "Don't have an account? Sign Up".

Figure 1 Login Screen Admin



The signup screen features a purple header and footer. The main content area is white and contains a purple circular icon with a white person silhouette. Below the icon, the text "Create an Account" is displayed in bold, followed by "Sign up to start using Personify". There are three input fields: "Email" with the placeholder "you@example.com", "Password" with a masked password ".....", and "Confirm Password" with a masked password ".....". A purple "Sign Up →" button is positioned below the confirm password field. At the bottom, there is a link that says "Already have an account? Log In".

Figure 2- Signup Screen Admin

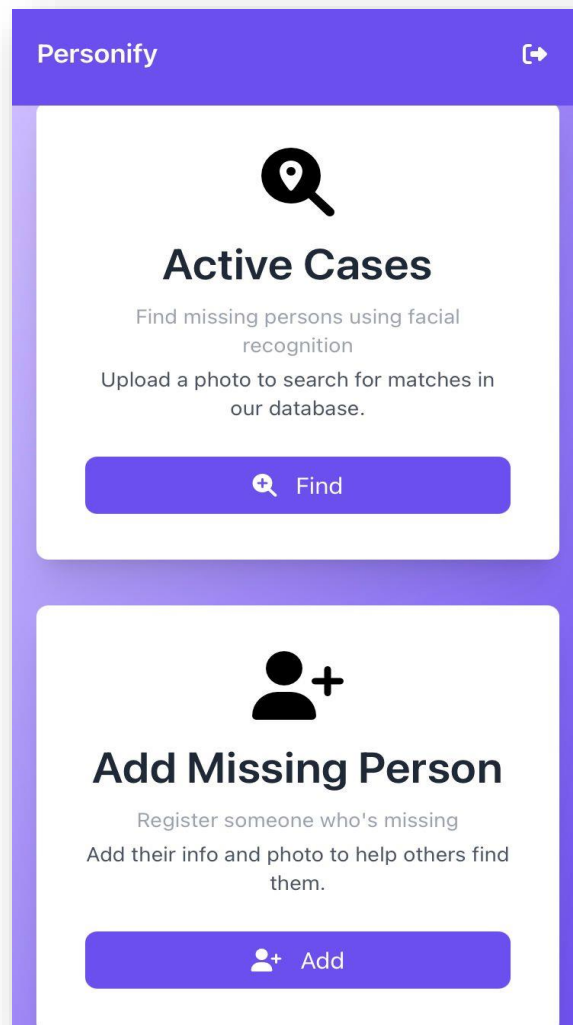


Figure 3- Dashboard Page

Add Missing Person

Please provide detailed information to help locate the missing person

Missing Person Details

 Upload Image

Drag and drop an image, or click to browse

Select Image

 Full Name

Enter full name of missing person

 Contact Information

Phone number or email

 Your Address (Reporter)

Enter your address

 Missing Place Address

Where was the person last seen

 Identity Details

Describe identifying marks, clothing, etc.

Submit Case

Figure 4- Missing Person Detail Page

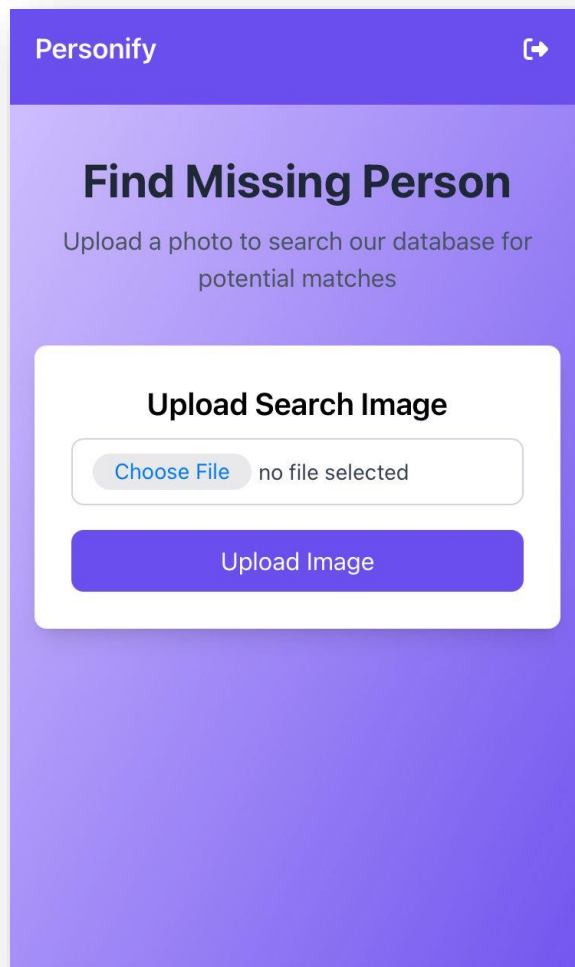


Figure 5- Find Missing Person page

4.7.2. Hardware Interfaces

- **Mobile devices:** A mobile app created by a company needs to communicate with smartphones and tablets to provide a usable experience for everybody
- **Servers and databases:** The application could need to contact servers and databases so it can add, remove and retrieve data from the database.
- **Cloud services:** When a user stores and processes data using cloud services, the application must access those services when required
- **Networking equipment:** In order to be connected to the internet and additional networks, the application can engage with routers and switches.

4.7.3. Software Interfaces

- Artificial intelligence can see images and videos by using algorithms within, so the data is recognized.
- In the case of a mobile app, it relies on frequent tools and frameworks in mobile app development.
- Should the application use cloud computing for important functions, it is important that it interacts with cloud services as required.

Following are the software components used for PERSONIFY android APP as show in table 3.7.3.

Table 2- 3.7.3 Software interface

Software Used	Description
Android Studio	In development of PERSONIFY APP android 5.0(Kit Kat) is used because of highest no's running android devices.
Deep Learning Algorithms	Deep learning is one of a group of modern machine learning techniques that use artificial neural networks for representation. You can choose between supervised, semi-supervised and unsupervised ways to learn.
DB	To generate a record of most searched items alongside user details.
Cloud Services	Firebase Real-time database and Firebase Storage is used.

4.7.4. Communications Interfaces

The medium for communication is the e-mail platform in this app anyone can build an account with the help of e-mail account and connect with our application. User will recover password if he/she forgot password.

4.8. System Features

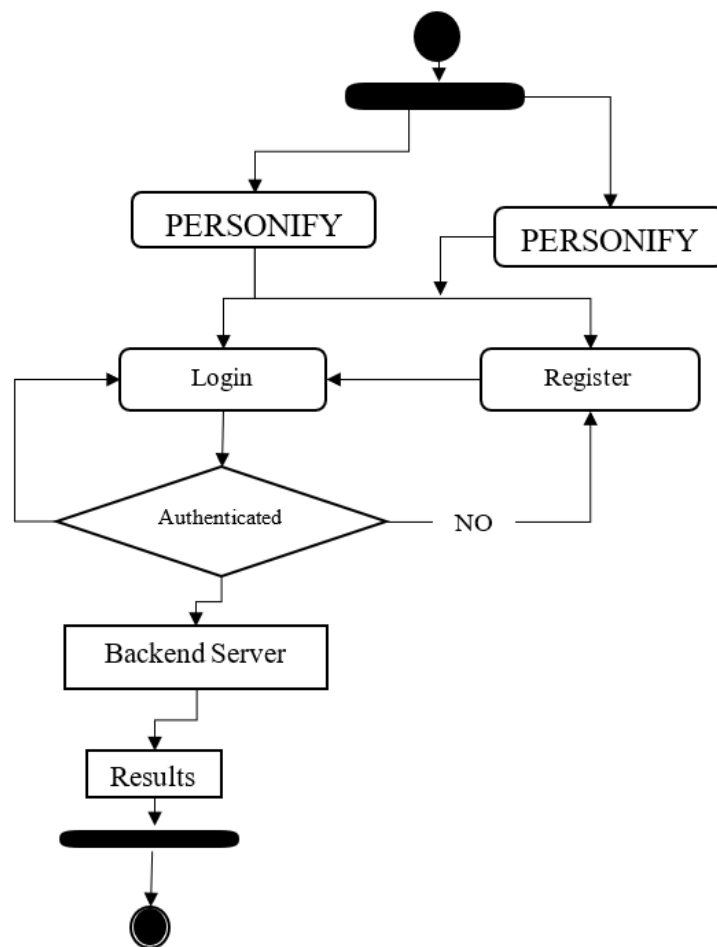
4.8.1. Admin Login and Selection of levels

USER CASE ID 01.

Use case number	UC-1
Use case name	Filling out form
Actor Name	Admin
Description	Database Entry
Preconditions	Must have internet and necessary detail of person with his image
Post conditions	None
Basic flow	Application → Interface → Data Entry → Record Entered

4.8.1.1. Description and Priority

Admin will login the PERSONIFY Admin app with his credentials. After logging into the app the admin will fill out a form and provide the persons image. Upon entering the required details the information will be uploaded to the server.



4.8.1.2. Response Sequences

- Upon entering the details the python script will be running which is the backend part of the application.
- The script will use the information to Detect faces and extract features.

4.8.1.3. Functional Requirements

- Data of the user will be secure fully.
- Allowing users to change their email.
- Password for admin.
- Admin can be viewing all data and information.

4.8.1.4. Client Login and Selection of levels

USER CASE ID 02.

Use case number	UC-2
Use case name	Filling out form
Actor Name	Client
Description	Image Entry
Preconditions	Must have internet and image of the person
Post conditions	Client will get the details and location upon match form the database
Basic flow	Application —————> Interface —————> Uploading image————> Person Details

4.8.1.1. Description and Priority

Client will login the PERSONIFY Client app with his credentials. After logging into the app the client will have to upload the image. Once the image is uploaded to the server. The client will get the record upon the matching description from within the database.

4.8.1.2. Response Sequences

- Upon entering the details the python script will be running which is the backend part of the application.
- The script will use the information to Detect faces and extract features.
- The client will get the record upon the matching description from within the database.

4.8.1.3. Functional Requirements

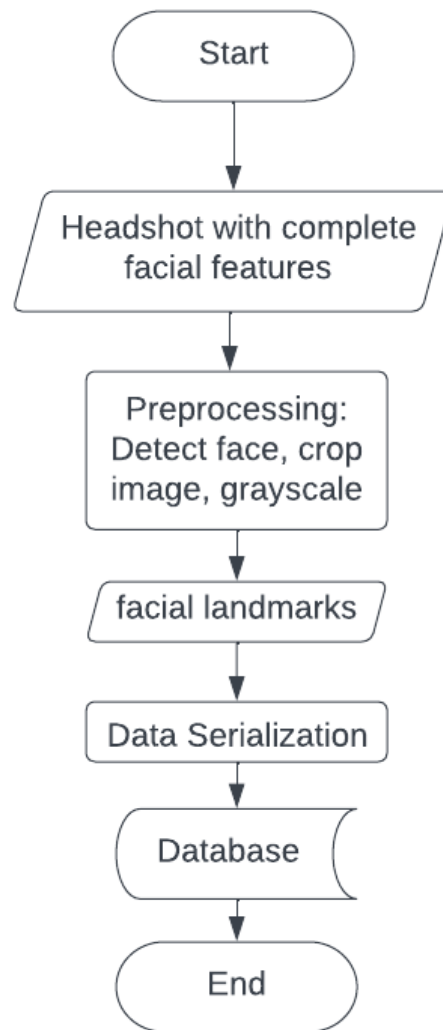
- Data of the user will be secure fully.
- Allowing users to change their email.
- Admin can be viewing all data and information.
- All the control will be given to the admin.

4.8.2. Product Information

The whole process requires a backend and a frontend. The necessary details are entered into the app. The Python face recognition algorithm extracts the features and matches along all the other facial features from the database. Upon matching provides the details.

4.8.2.1. Description and Priority

Two important aspects of responsible pet ownership are to give animals access to drinkable water that is cool and good food at all times. Feeding your pet a good diet will add joy to its life. In short, the user will find all the details of the app on the app page. This feature is currently set as priority 09.



4.8.8.2. Response Sequences

- Automatically run the Feature software program.
- Detect faces: The application will first detect all the faces present in the input image or video stream.
- Align faces: Once the faces are detected, the application will align them to a standardized pose using Dlib's facial landmark detection algorithm. This step involves identifying key points on the face, such as the eyes, nose, and mouth, and rotating the face so that these points are in a fixed position.

- **Extract features:** After aligning the faces, the application will extract a set of features from each face using Dlib's face recognition model. These features represent unique characteristics of the face, such as the shape of the eyes or the distance between the nose and mouth.
- **Compare faces:** Once the features have been extracted, the application can compare them to a database of known faces to identify the person in the input image. This step involves calculating the distance between the feature vectors of the input face and each face in the database, and selecting the closest match.
- **Return result:** Finally, the application will return the result of the face recognition process, which may include the identity of the person in the input image, a confidence score indicating how likely the match is correct, or other information such as the location of the face in the image.

4.8.2.3. Functional Requirements

- Supplying users with a product-based collection of marts.
- The application will detect faces through the images provided.
- The process would involve facial landmark detection such as eyes, nose, mouth and chin.
- Feature extraction will be performed after that. Which involves calculating distances between the eyes, width of the nose among other things, and use them to create a unique facial signatures.

4.9. Other Nonfunctional Requirements

4.9.1. Performance Requirements

The application should be light weighted for the productive response in case of any query put forward by the user. The proportion of the throughput of the application should be at the peak.

- **Accuracy:** The application will have a high level of accuracy in identifying matches between images and missing persons.
- **Efficiency:** The application will be able to process large amounts of data quickly and efficiently in order to locate missing persons in a timely manner.

- **Scalability:** The application will be able to handle a large volume of data and users without degrading performance.
- **Security:** The application will be secure and protect sensitive data, such as user profiles and information about missing persons, from unauthorized access and tampering.
- **Privacy:** The application will respect user privacy and comply with relevant laws and regulations, such as data protection requirements.
- **User experience:** The application will be easy to use and provide a seamless and intuitive experience for users.

4.9.2. Safety Requirements

The application will ensure the proper safeguard of the personal details and privacy of the users. There will also be a backup phenomenon in case of any deterioration or loss of data.

- **Data integrity:** The application should ensure the integrity and reliability of data about missing persons and user profiles.
- **Data backup and recovery:** The application should have measures in place to protect data from loss or damage, and to allow for recovery in the event of a failure or disaster.
- **Risk assessment and management:** The application should assess and manage potential risks to the security and privacy of data, as well as to the safety of users and missing persons.
- **Compliance with laws and regulations:** The application should comply with relevant laws and regulations, such as data protection and privacy laws, in order to protect the safety and rights of users and missing persons.
- **Emergency response:** The application should have measures in place to respond to emergencies and provide assistance to users and missing persons as needed.

4.9.3. Security Requirements

Users will be equipped with the protection from unwanted criticism. Data base backup provided in case of malfunction, Application protection to avoid attacks. User's privacy is our first priority. Some questions will be asked from the user when making their accounts and user must give the answers to these questions. If customer will be unable to access the data, then it is necessary to ask these questions to properly recover the account.

4.9.4. Software Quality Attributes

4.9.4.1. Internet Protocols

Android phone must meet with the requirements of the TCP/IP protocol and must configure accordingly

4.9.4.2. Information Security Requirement

International standards for information protection are supported by the scheme.

4.9.5. Business Rules

4.9.5.1. Maintenance

The maintenance of the system is conducted in compliance with the maintenance contract.

4.9.5.2. Standards

As per the IEEE guidelines, the coding standards and naming conventions will be.

4.9.5.3. Licensing Requirements

Use is limited to those who sign up for the account and demonstrate that they comply with the CCA rules by using them.

4.9.5.4. Legal, Copyright and other Notices

The PERSONIFY App is a private product of that product's developers and no one can be use the app without admin's permission. Conflicts will be dealt with under the Judaism of Islamic Republic Pakistan. According to the statute, the legal authority of this software will be held liable for any copyright disputes and must settle the patent issues. If any anomaly arises in the processing or accession of the database inquiry and obtaining the user's information, it must be assured that it will be done professionally according to the guidelines set out in the agreement between the users.

Methodology

5.1. Introduction

The purpose of the process is to reach the aim of the project, with a perfect outcome. To assess this project, a method known as System Development Life Cycle (SDLC) follows three major stages, as represented in Fig 7: Implementing, planning and analysis.

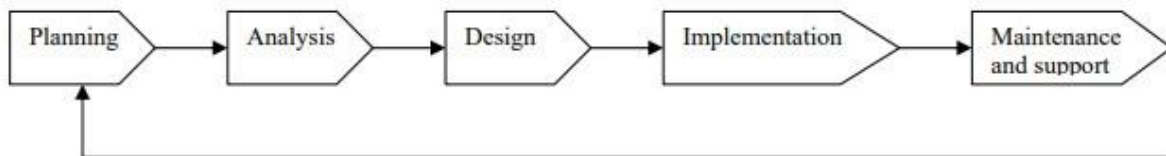


Figure 6- SDLC

This project use three major steps to implement project starting with Planning, Implementing and testing. All these steps use for finding, analyzing of data according to the meet of project.

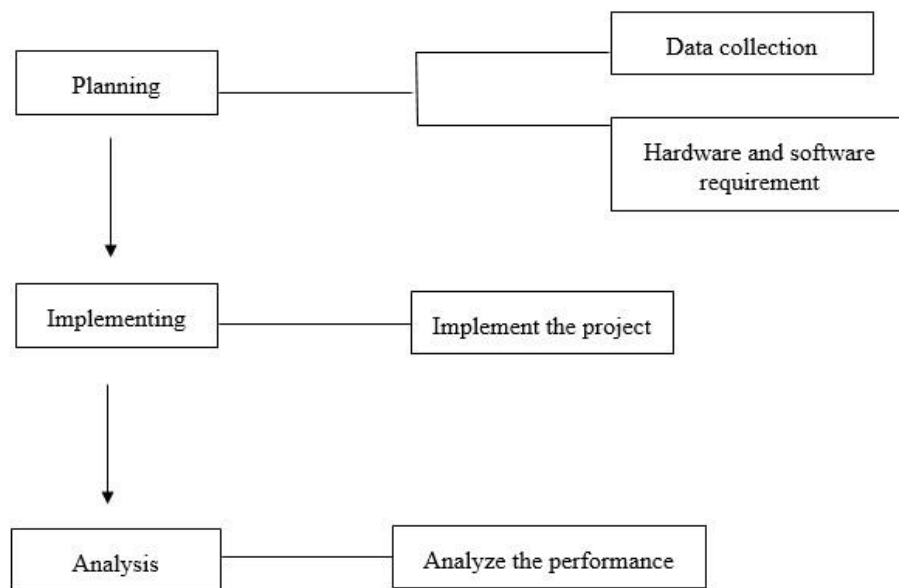


Figure 7- Steps of methodology

5.2. Planning

Before handling hardware and software needs, the planning stage should cover all important project information. There are two main parts of Explain CM: gathering data and meeting the needs for hardware and software.

5.2.1. Data Collection

Gathering data is a key part of all projects. It was in that stage I intended to gather the requirements and obtain all the necessary information through the internet, researches and books. After gathering the data from different sources, important pieces are chosen to guide the development of the project

5.2.2. Hardware and software requirements

The hardware and software requirements are as follow.

5.2.2.1. Hardware requirements

Smart Phone Sim supported (having android greater than 5.0) and a windows operated system with python environment installed.

5.2.2.2. Software requirements

- Android Studio
- Visual Studio Code

5.3. Implementing

Now, all the gathered data from the first phase is used with software to build the App. When the app requirements are complete and properly designed for the main aim, you go ahead to the end phase.

5.4. Analysis

When all the right data is included in your project, the following step is to confirm the appropriate use of project methodologies.

Testing completes the project requirements by the Analysis phase's end. The first step for any user is to sign up and after that, log in to use the application. When a user selects to save their password, they will be logged in automatically in future. It is easy for anyone to log in to the platform. It will be very easy for anyone to use.

Detailed Design and Architecture

6.1. System Architecture

Our application PERSONIFY is built using the agile model. Ensure the following information is different when using a complete native application. Applications are created in the frontend and a layer of python brings in machine learning to work with the application's data. An Android-based client app will talk to a backend server that is on the network. It will process the image by identifying it with facial landmarks and using facial recognition. All face templates and up-to-date identity results of users are stored and overseen by the database.

An Android app refers to the part of the application users participate with, located on the frontend. Android's version of the app will take images, place them on the back end and show the user their results.

The app sends pictures to the backend server which then locates facial landmarks using dlib and sends the results to the app. The backend server will also keep track of the templates used and all of a user's recognition history.

Every user's identifier information such as their face templates and data from recognitions, will be securely kept in the database. The backend server will look after both receiving and saving the data, interacting with the database as needed.

Detection of facial landmarks in the system is handled by the dlib library for facial landmark detection. It modifies and takes out important features from the input image for easier identification.

Facial features from the input photo will be compared by the system to faces already included in the database. After the recognition engine sends the result to the backend server, the server gives it back to the Android application.

6.1.1. Architecture Design Approach

For the development of PERSONIFY App Architecture design are as follow.

- Initial Approach.
- Test.
- Next initial.

6.1.1.1. Initial Approach

- Design the initial technique of current problem.
- Create the module one by one.
- Test design after module completion.

6.1.1.2. Test

- Environmental test (runs on condition to meet the real time response).
- Module run on different testing techniques.

6.1.1.3. Next Initial

- Repeat the first two step.
- Terminate.

6.1.2. Architecture Design

The System Architecture Design is given Figure 9.

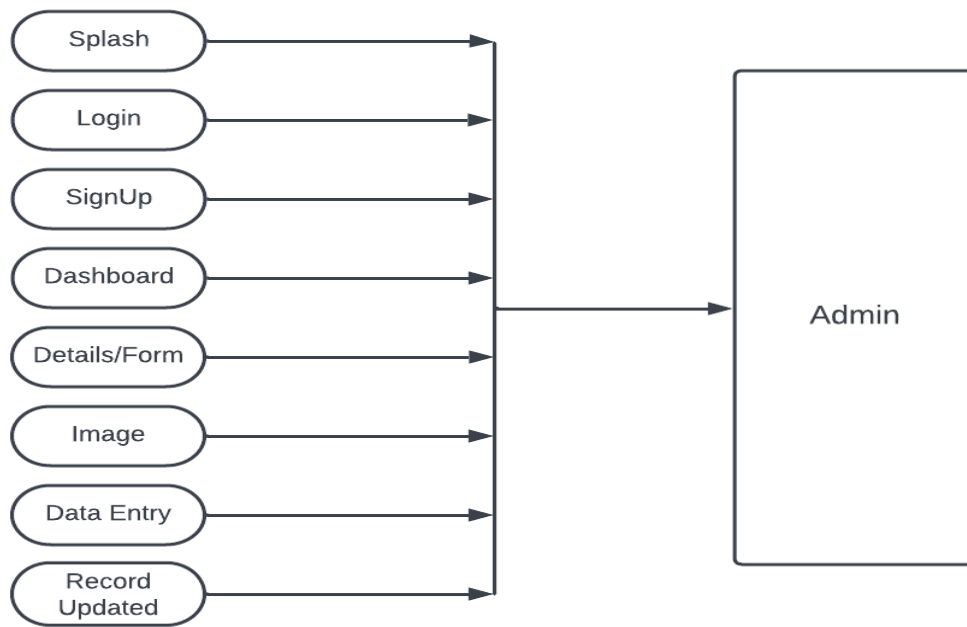


Figure 8- System Architecture Design Admin

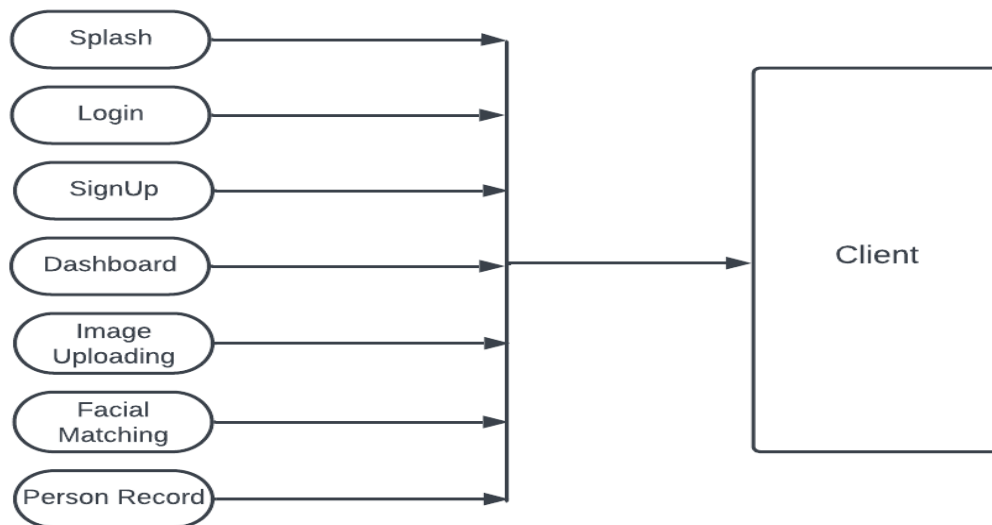


Figure 9- System Architecture Design Admin

6.1.3. Subsystem Architecture

Subsystem Architecture are described with Functional description.

6.1.3.1. Functional Description

All modules in the PERSONIFY App are described, along with a larger overview of the app, as well as small overviews for every individual module which are displayed in Fig 10 and Fig 11.

- Login/ Signup.
- Person Details.
- Matching Facial features.
- Displaying Record.

6.2. DETAILED SYSTEM DESING

6.2.1. Classification

- Login/ Signup.
- Data/Details Entry.
- Features extraction of Images.
- Client uploading image.
- Algorithm matching features with the ones in database.
- Displaying Record.

6.2.2. Definition

Those with the authority in the system are named admins and look after the database. A client is any person who wants help by searching for a missing individual or reporting an offender.

6.2.3. Responsibilities

With PERSONIFY, the Admin can select and capture videos or images directly from the device's camera.

Get them from other sources. You should see images or videos you've captured on the screen.

the app allows users to check the captured images or videos. The frontend needs to be made with the user's experience in mind.

A way for the user to use the app such as starting the face recognition process.

The frontend takes captured images or videos and gives them to the backend to handle.

The images or videos sent by the frontend need to be received by the PERSONIFY Client App. processing. It's important for the backend to detect faces using the dlib facial landmark detection features.

The apps connects to a server outside and receives the images or videos that were sent from the frontend. The system on the backend should be able to find the facial features.

both the faces you detect and where they place the eyes, nose and mouth. The backend ought to analyze the detected faces against a collection of facial pictures a photograph or illustration that shows the person's face. Providing the results of face analysis is the main task of the backend.

the recognition occurs in the frontend. The facial recognition data needed should be stored in the database.

The app helps in performing the face recognition. It is important that the database allows facial recognition to be carried out.

When the face is processed, data is sent to the backend for a comparison.

6.2.4. Constraints

- Being responsible for user data is important because the app uses that information. It is important to design the app so that nobody can look at or use user data without the user's permission.
- Face recognition becomes very challenging when trying to process large amounts of data. Your app should manage a lot of data effectively, using only a limited amount of resources.
- Creating a face recognition app on Android with dlib for landmark detection is not cheap. Cost and value should line up in the design process for building and using the app.

6.2.5. Composition

There is no subcomponent of this Component.

6.2.6. Uses/interaction

First records have to be entered using PERSONIFY Admin App into the database. Afterwards, by using PERSONIFY Client App missing/criminal persons could be identified using the app.

6.2.7. Resources

- This app was built using the help of Android Studio.
- The frontend of an Android app should be built with Java and Python and the dlib library's facial landmark detection algorithms can be used for the backend.
- It's necessary to include a powerful CPU, enough memory and lots of storage for designing and testing web applications.
- App developers can use Firebase Real-time database and storage for storing the app's data.

6.2.8. Processing

Our app users must be able to access extensive information about popular products studied in our system and further select the ones related to common work patterns within the industry.

6.2.9. Interface/Export

It provides the following services.

- Registration
- Login
- Forgot password.
- Person Details Form
- Person Identifier
- Person Details
- Log out

6.2.10. Detailed Subsystem Design

There is no Subsystem Design for this Project.

Diagrams

1. Use case Diagram.
2. Architectural Diagram.
3. ER Diagram.
4. Activity Diagram.
5. Sequence Diagram.
6. Data Flow Diagram.

6.2.10.0. Use case Diagram

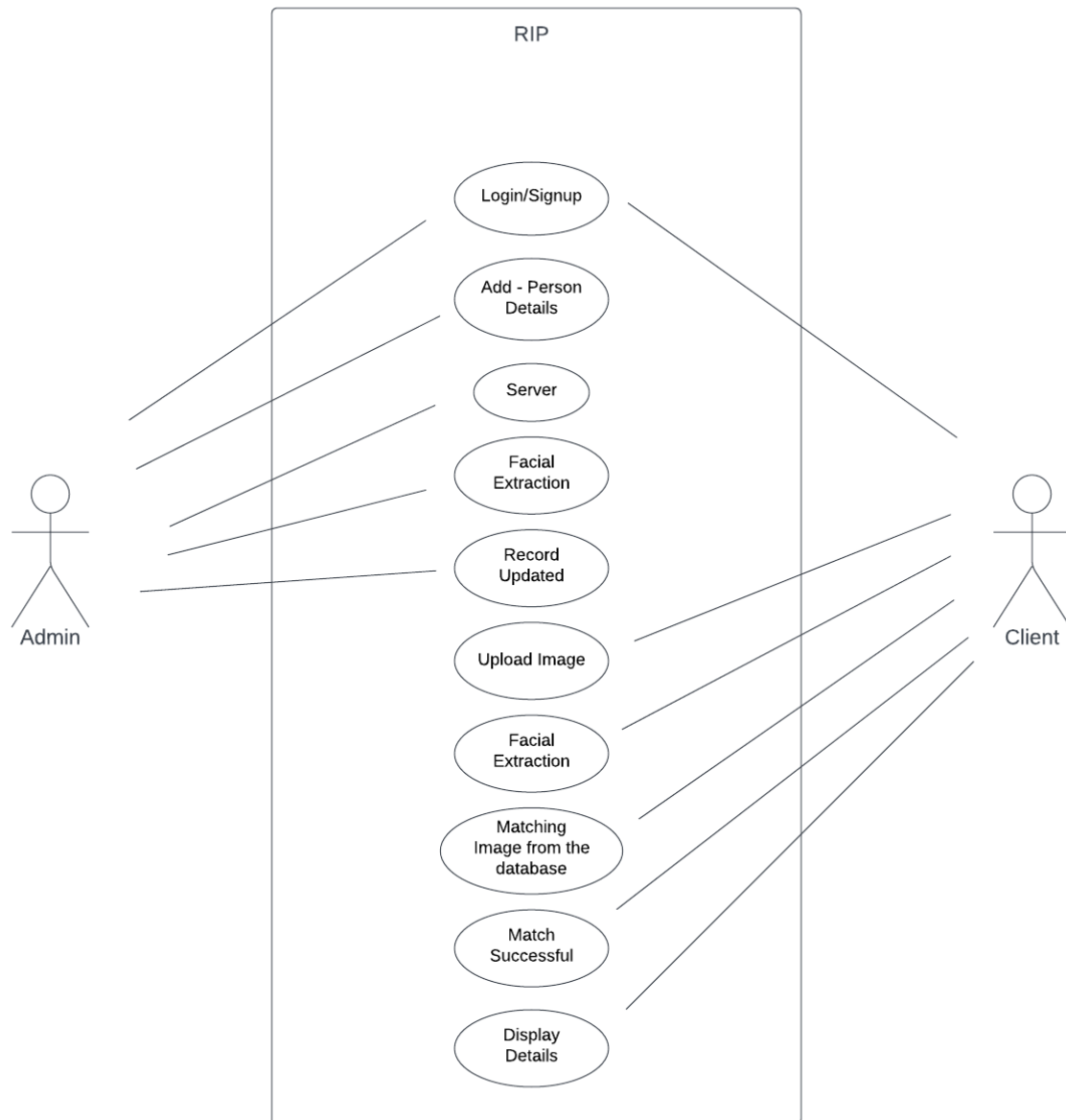


Figure 10- Use case Diagram

6.2.10.1.1 Use Case: Admin App User login/ signup

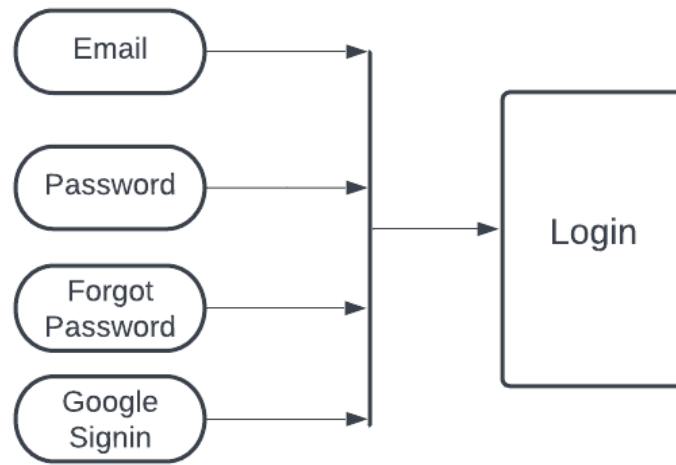


Figure 11- Use Case

Description: The following use case is briefly explained in chap 3 SRS.

6.2.10.1.1.1 Use Case: Admin App Person Details

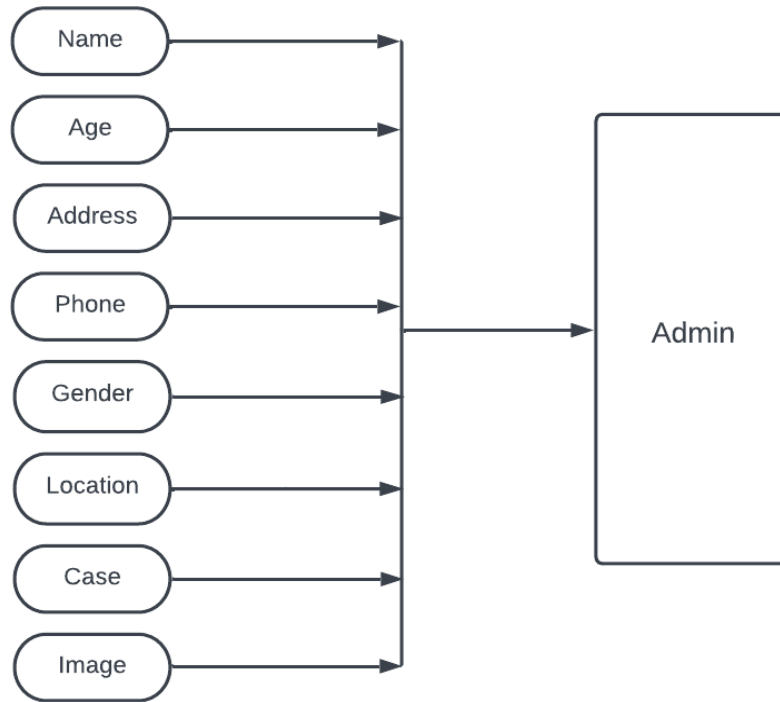


Figure 12- Use Case Admin App

6.2.10.1.1.2 Use Case: Client App User login/ signup

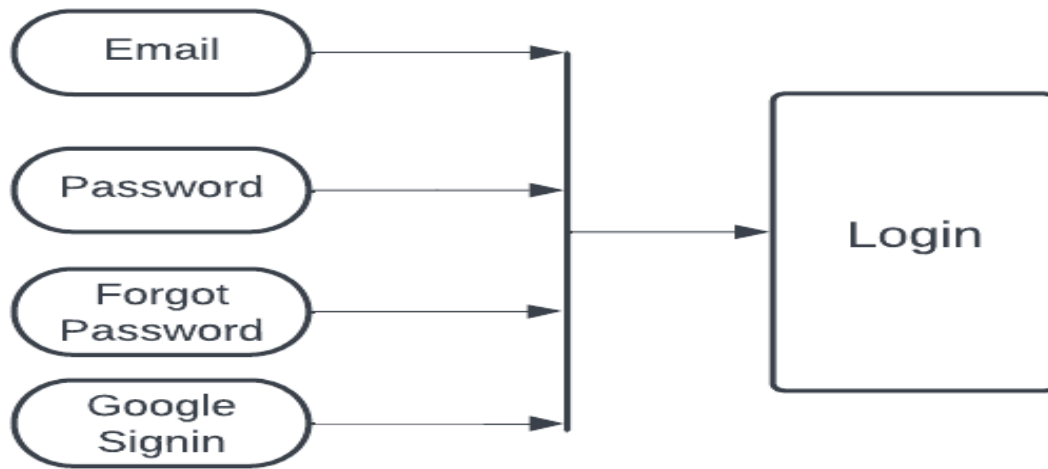


Figure 13- Use case Home screen

Description: The following use case is briefly explained in chap 3 SRS.

6.2.10.1.1.3. Use Case: Client App Dashboard

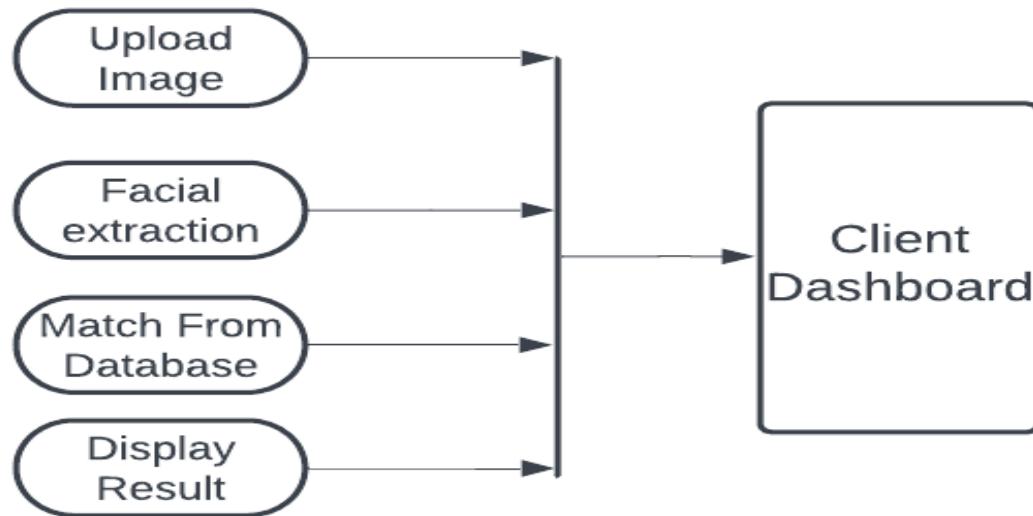


Figure 14- Use Case Client

6.2.10.2. Activity Diagram

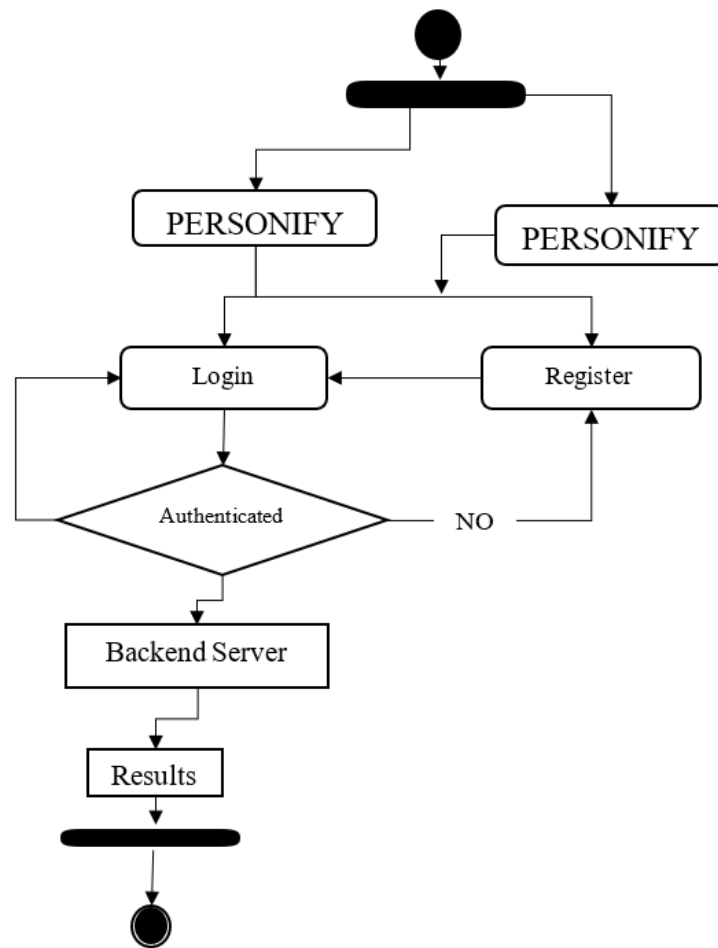


Figure 15- Activity Diagram

6.2.10.5. ER Diagram

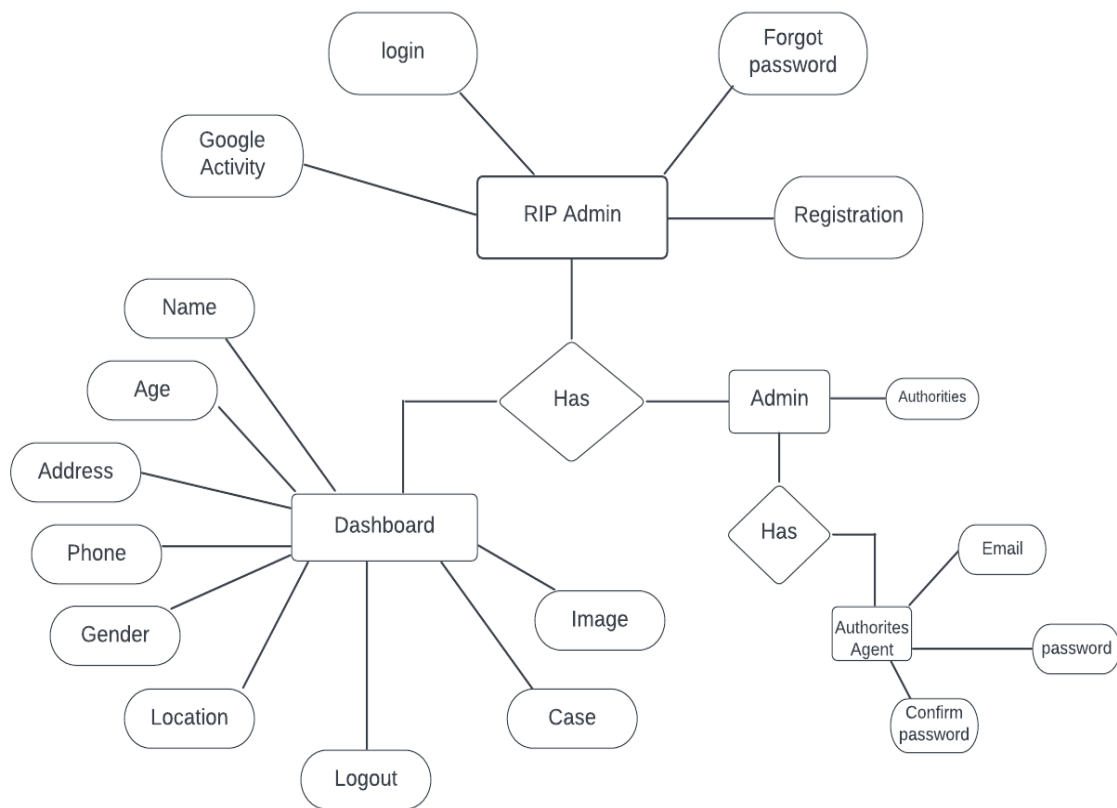


Figure 16-(a) ER-Diagram Admin

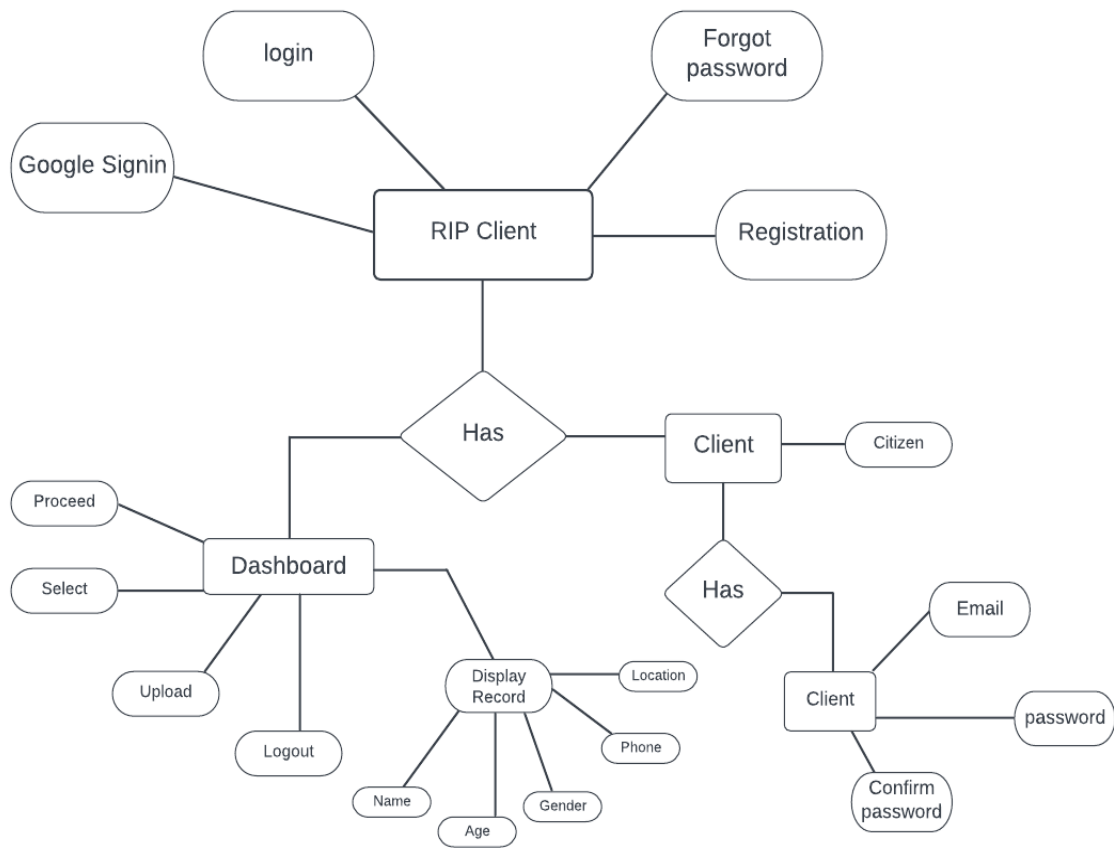


Figure 17- (b) - ER-Diagram Client

6.2.10.6. Architectural Diagram

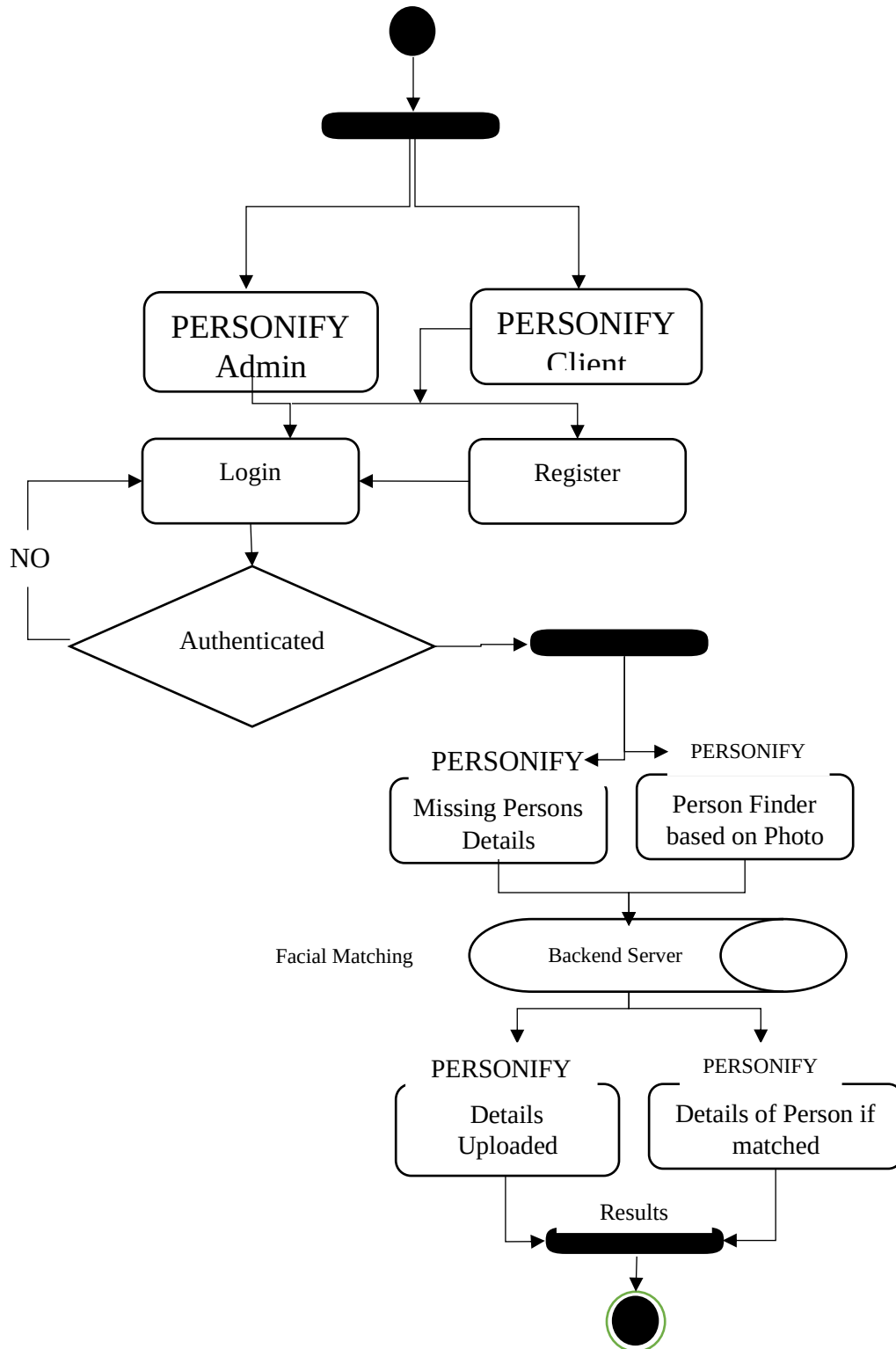


Figure-18-Architectural Diagram

6.2.10.7. Data Flow Diagram

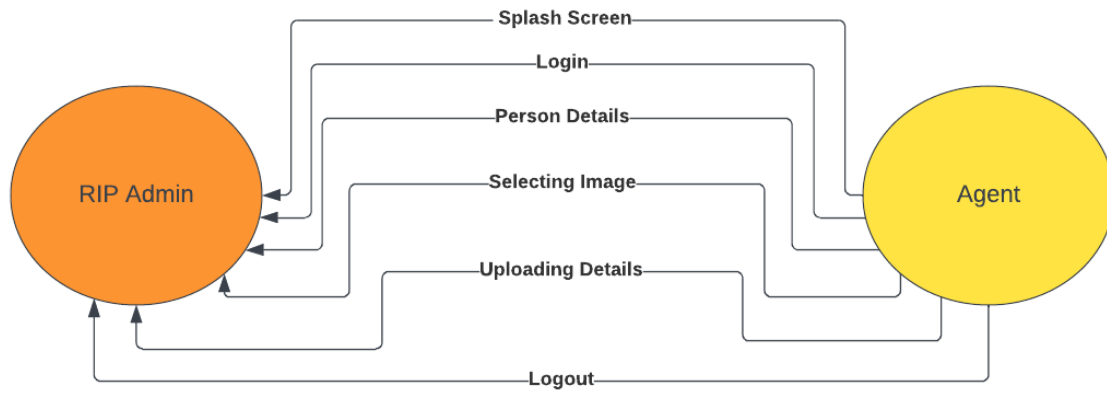


Figure 19- DFD Admin

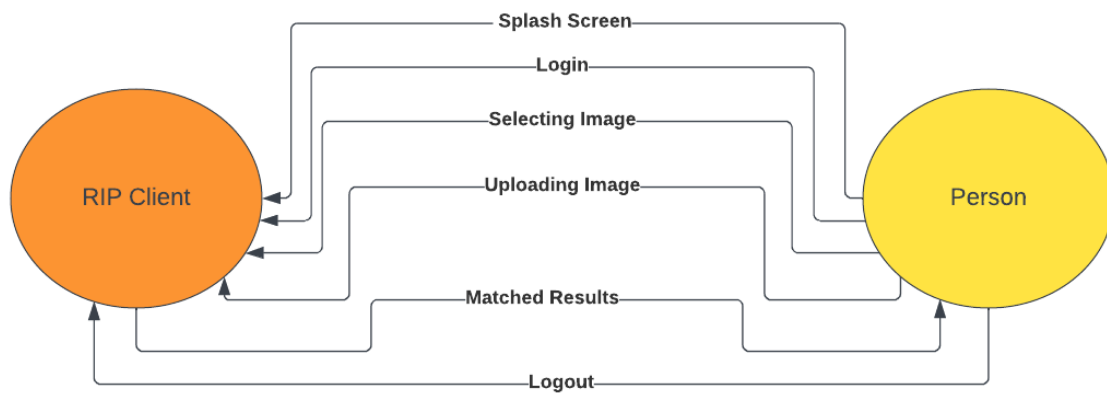


Figure 20- DFD Client

Chapter 7

Implementation and Testing

7.1 Tools

- Android Studio 3.4.1.
- Visual Studio Code.
- Smart Phone (Having Internet).

7.2 Testing methods

In this process we test two main Activities of Login Screen Admin as well as Client.

7.2.1 Test plan for Login/ Signup Screen.

Screen Name: Login/Signup.

Test Plan ID: 001

APP Name: **Personify App.**

Engineers: **Muhammad Adeel / Munawar Ihsan**

Test no: 1

Purpose

Checking the APP activity working Login/Signup or not.

Scenario:

APP is responsive to all devices having android version 6.0(marshmallow) or greater and also to complete that module to cover Dimensions.

Environment:

Android Studio.

Pre-Requisite:

Observing / Checking Login Signup.

Strategy:

- Click on “Sign in” Section to open required Screen.
- Fill out all the information required for it.
- Enter the information all necessary.
- Show the profile of user.

Expected Result:

- Selected screen display properly and achieved result.
- All section are working properly.

Observation:

- All section are selected and working successfully.
- Login Screen Works Successfully.

7.2.2 Test plan for Checking Features.

Screen Name: Record Display.

Test Plan ID: 002

APP Name: **Personify App.**

Engineers: **Muhammad Adeel / Munawar Ihsan**

Test no: 02

Purpose

Checking if an image uploaded matches from the database and the record is displayed.

Scenario:

APP is responsive to all settings that enabled

Environment:

Android Studio/Smart Phone.

Pre-Requisite:

Observing / Features.

Strategy:

- Click on “Details Button” Section to open required activity.
- Show all Record.

Expected Result:

- Selected screen display properly and achieved result.
- Record is being displayed.

Observation:

- If the image uploaded to the server matches from the database, record is being displayed.

Results and Discussion

8.1 Results of Test Evaluation

From the previous chapter testing evaluation results are shown in figure 7.1(a, b), 7.2(a, b).

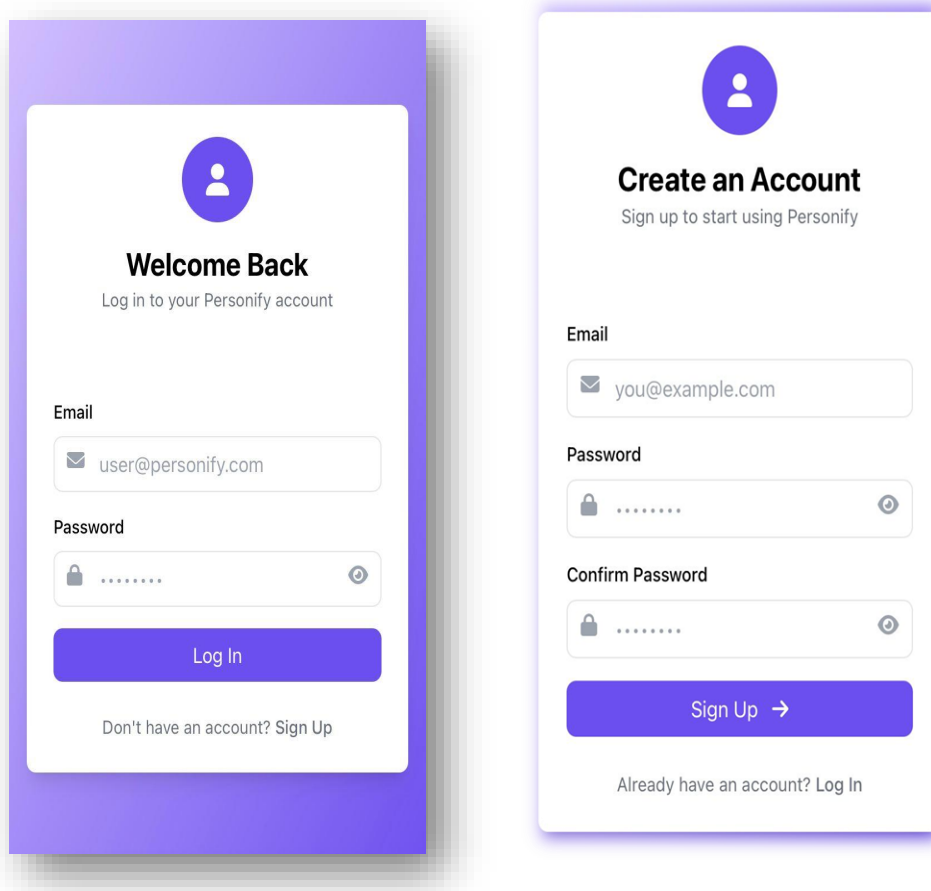


Figure 7.1(a) – Figure 1- Login Screen Admin 7.1 (b) – Signup Screen Admin

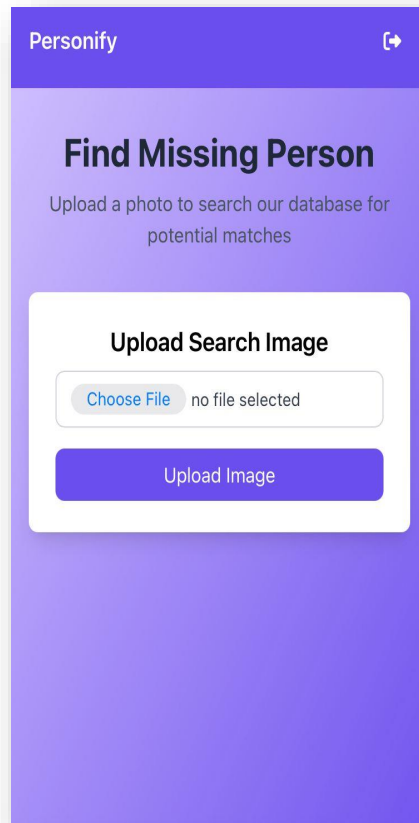


Figure 7.1(c) – Find Missing Person Page

All the details regarding these images were completely described in chapter 3 and 5.

Conclusion and Future Work

9.1 Future Work

PERSONIFY app is the best solution for most of the problems that the user faces. Our first priority is to ensure the system is secure such that the user will feel safe using it. Other user's data will be safe. Face recognition is an ever-changing technology and there can always be the likelihood to make the app more accurate by integrating other new algorithms or methods. User experience plays a huge part in whether an app succeeds or fails. Developers can improve usability of the app by updating the UI, making the app to perform better and responding to user feedback. The source code will be available at some point, but developers are allowed to develop new functions and features for it.

9.2 Conclusion

As of now, the application "PERSONIFY" that we have developed is operating in a certain region. This restriction is both purposeful and beneficial in providing a scope for observing, testing, and refining the system. When tested on a smaller scale, sensitive data, private user information, and data privacy are protected, spying free user hands-on experience access. Trial phase bottom monitoring operate app it able conditions real world

Gaining actual user data assists in resolving issues relating to reliability, precision, and overall performance. Gaining appraising user experience fulfilled defined objectives at outlined targets has also empowered collecting gauging satisfaction gauging. This feedback steers further modifications directed to enabling defined goals functional goals with the objectives. Privacy concerns, legal responsibilities, and technical challenges the need addressing scope gradually needs managed with wider audience availability eliminating enable growth first app's expanding.

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